

ANDONG, Korea, Republic of

WMO: 471360

Lat: 36.5731N

Lon: 128.7072E

Elev: 142

StdP: 99.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-9.2	-22.3	0.5	-5.4	-20.1	0.6	-4.6	8.0	-2.7	7.0	-2.2	1.6	290	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	33.1	24.1	31.8	23.6	30.5	23.0	25.7	30.1	25.2	29.4	24.7	28.6	2.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.1	27.6	24.1	19.3	27.2	23.5	18.6	26.9	80.5	30.3	78.2	29.5	75.9	28.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
6.2	5.4	4.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-14.4	34.9	2.2	1.4	-16.0	35.9	-17.2	36.7	-18.5	37.5	-20.1	38.6		
			-15.2	26.5	2.1	1.1	-16.8	27.3	-18.0	28.0	-19.2	28.6	-20.7	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-1.6	1.0	6.2	12.5	17.9	22.0	25.2	25.6	20.5	14.4	7.1	0.4		
	DBStd	9.96	3.49	3.58	3.61	3.55	2.69	2.32	2.64	2.74	2.78	3.26	3.90	3.63		
	HDD10.0	1159	358	252	127	15	0	0	0	0	0	6	103	298		
	HDD18.3	2728	617	485	377	178	40	2	0	0	10	126	337	557		
	CDD10.0	2131	0	1	8	89	246	359	471	483	316	142	16	0		
	CDD18.3	658	0	0	0	2	27	112	213	224	76	4	0	0		
	CDH23.3	5417	0	0	1	51	331	830	1806	1982	402	14	0	0		
	CDH26.7	1767	0	0	0	6	65	232	647	743	74	0	0	0		
(14) Wind	WSAvg	1.6	1.9	1.9	2.1	2.0	1.7	1.6	1.4	1.4	1.3	1.2	1.4	1.7		
(15) Precipitation	PrecAvg	1024	21	34	45	81	87	120	220	183	130	41	35	18		
	PrecMax	1334	91	116	92	181	235	262	331	354	297	116	91	42		
	PrecMin	709	0	4	4	26	29	41	56	57	15	3	3	1		
	PrecStd	202	27	30	23	43	57	56	86	90	86	35	28	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	14.8	20.8	27.2	30.6	32.3	34.7	35.3	31.0	25.0	20.0	12.2		
		MCWB	4.8	9.2	10.7	15.1	17.1	19.7	24.7	24.6	22.9	17.2	13.4	7.4		
	2%	DB	7.4	11.6	18.0	24.8	28.4	30.7	33.1	33.5	28.7	23.2	17.2	9.7		
		MCWB	3.3	5.6	9.2	13.5	16.3	19.7	24.6	24.4	21.2	16.1	11.5	5.3		
	5%	DB	5.7	9.4	15.7	22.6	26.8	29.3	31.7	32.2	27.1	21.7	15.4	8.0		
		MCWB	2.0	4.1	7.8	12.3	15.8	19.4	24.2	24.4	20.4	15.5	10.8	4.3		
	10%	DB	4.2	7.4	13.5	20.4	25.1	27.8	30.2	30.7	25.8	20.2	13.5	6.3		
		MCWB	0.8	2.9	7.1	11.5	15.3	19.3	23.9	24.1	19.9	14.9	9.6	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	10.3	12.9	18.4	20.9	24.3	26.7	26.7	24.7	20.0	15.5	8.8		
		MCDB	8.2	13.4	16.7	23.1	24.6	27.4	31.3	31.0	28.6	22.0	17.7	11.1		
	2%	WB	4.2	6.5	10.8	15.4	19.1	22.9	25.7	25.9	23.5	18.3	13.2	6.5		
		MCDB	6.6	10.1	15.6	21.6	24.0	26.1	30.3	30.2	26.5	20.9	15.8	8.9		
	5%	WB	2.5	4.8	9.0	13.8	17.8	21.8	25.2	25.4	22.2	17.0	11.6	4.8		
		MCDB	5.0	8.6	14.3	19.9	23.3	25.9	29.5	29.6	25.1	20.0	14.4	7.3		
	10%	WB	1.2	3.3	7.5	12.5	16.7	20.8	24.7	24.9	20.9	15.6	10.1	3.4		
		MCDB	3.7	7.0	12.4	18.1	22.4	25.1	28.7	29.0	24.1	19.1	13.0	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.6	11.4	12.6	12.2	10.1	7.5	7.9	9.1	10.9	10.6	9.8		
		MCDBR	11.3	13.5	15.4	16.7	16.0	13.4	9.9	10.2	10.8	12.0	12.8	11.6		
	5% WB	MCWBR	7.6	8.2	7.9	6.9	5.8	4.4	3.0	3.0	3.9	5.4	7.3	7.5		
		MCWBR	9.8	11.7	12.7	13.1	11.7	8.7	8.3	8.4	8.0	8.6	10.5	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.399	0.452	0.541	0.574	0.583	0.634	0.596	0.545	0.473	0.452	0.425	0.392		
		taud	2.038	1.933	1.762	1.743	1.765	1.718	1.875	2.007	2.149	2.075	2.067	2.079		
	Ebn at Noon		752	756	723	729	731	693	716	745	781	750	718	729		
		Edh at Noon	128	161	210	225	224	234	199	171	141	138	124	116		
(44) All-Sky Solar Radiation	RadAvg		2.62	3.34	4.30	4.95	5.57	5.13	4.37	4.36	3.88	3.53	2.60	2.34		
	RadStd		0.17	0.34	0.42	0.37	0.50	0.39	0.58	0.56	0.45	0.31	0.33	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	N/A

Nomenclature: See separate page